高等微积分(2)

第十四章: 多变量函数的微分学

14.8　反函数定理

问题的提出: 向量值函数 $\vec{g}:\Omega\subset\mathbb{R}^n\to\mathbb{R}^n$ 是否能有反函数 $\vec{g}^{-1}$?

这等价于问方程 $X=\vec{g}(Y)$ 是否有解 $Y=\vec{g}^{-1}(X)$, 也即方程

$$F(X,Y):=\vec{g}(Y)-X=\vec{0}$$

是否有隐函数解 $Y=\vec{g}^{-1}(X)$?

向量值函数 $X=\vec{g}(Y):\Omega\subset\mathbb{R}^n\to\mathbb{R}^n$ 可以写成

$$X=(x_1,x_2,\dots,x_n)=\Big(g_1,g_2,\dots,g_n\Big)\Big|_Y,$$

这里

$$Y=(y_1,y_2,\dots,y_n)\in\Omega.$$

$$x_j=g_j=g_j(Y)=g_j(y_1,y_2,\dots,y_n)\in\mathbb{R},$$

$$j=1,2,\dots,n.$$

定理 1. 设 $k \geqslant 1$ 为整数, $\Omega \subset \mathbb{R}^n$ 为非空开集, $Y_0 \in \Omega$, $\vec{g}: \Omega \to \mathbb{R}^n$ 为 $\mathcal{C}^{(k)}$ 类使得 $J_{\vec{g}}(Y_0)$ 可逆. 令 $X_0 = \vec{g}(Y_0)$. 则 $\exists \delta, \eta > 0$ 使得 $B(Y_0, \eta) \subset \Omega$, 且存在 $\vec{f}: B(X_0, \delta) \to B(Y_0, \eta)$ 为 $\mathcal{C}^{(k)}$ 类使得 $\forall X \in B(X_0, \delta), \forall Y \in B(Y_0, \eta)$, 等式 $X = \vec{g}(Y)$ 成立当且仅当 $Y = \vec{f}(X)$. 另外, $\forall X \in B(X_0, \delta)$,

$$J_{\vec{f}}(X) = \left(J_{\vec{g}}(\vec{f}(X))\right)^{-1} = \left(J_{\vec{g}}(Y)\right)^{-1}.$$

即 $\vec{g}$ 在点 Y_0 处具有局部的反函数(局部可逆).

证明: 选取 $r>0$ 使得 $B((X_0,Y_0);r)\subset\mathbb{R}^n\times\Omega$.

$\forall(X,Y)\in B((X_0,Y_0);r)$, 定义

$$\vec{F}(X,Y)=\vec{g}(Y)-X.$$

则 $\vec{F}$ 为 $\mathcal{C}^{(k)}$ 类. 记 $\vec{F}=(F_1,\ldots,F_n)^T$, 那么

$$\frac{\partial(F_1,\ldots,F_n)}{\partial(y_1,\ldots,y_n)}(X_0,Y_0)=J_{\vec{g}}(Y_0)$$

为可逆矩阵. 由隐函数定理知, $\exists\delta,\eta>0$ 使得

$$B(X_0,\delta)\times B(Y_0,\eta)\subset B((X_0,Y_0);r),$$

且 $\forall X \in B(X_0,\delta)$, $\exists! Y \in B(Y_0,\eta)$ 使得我们有 $\vec{F}(X,Y)=0$. 令 $\vec{f}(X)=Y$, 那么 $\vec{g}(\vec{f}(X))=X$, 并且 $\vec{f}: B(X_0,\delta)\to B(Y_0,\eta)$ 为 $\mathcal{C}^{(k)}$ 类向量值函数使得 $\forall X \in B(X_0,\delta)$, 我们均有

$$\begin{aligned} J_{\vec{f}}(X) &= -\Big(\frac{\partial(F_1,\ldots,F_n)}{\partial(y_1,\ldots,y_n)}(X,\vec{f}(X))\Big)^{-1} \\ &\quad \cdot\frac{\partial(F_1,\ldots,F_n)}{\partial(x_1,\ldots,x_n)}(X,\vec{f}(X)) \\ &= \Big(J_{\vec{g}}(\vec{f}(X))\Big)^{-1}. \end{aligned}$$

例 1. (极坐标变换) 令 $D=(0,+\infty)\times(-\pi,\pi)$.

$\forall(\rho,\varphi)\in D$, 定义

$$\vec{f}(\rho,\varphi)=\begin{pmatrix} x \\ y \end{pmatrix}=\begin{pmatrix} \rho\cos\varphi \\ \rho\sin\varphi \end{pmatrix}.$$

则 $\vec{f}$ 为 $\mathcal{C}^{(\infty)}$ 类向量值函数且

$$J_{\vec{f}}(\rho,\varphi)=\begin{pmatrix} \cos\varphi & -\rho\sin\varphi \\ \sin\varphi & \rho\cos\varphi \end{pmatrix},$$

从而 Jacobi 行列式 $\det J_{\vec{f}}(\rho,\varphi) = \rho > 0$. 于是 $\vec{f}$ 为局部可逆, 其逆映射 $\vec{f}^{-1}$ 也为 $\mathcal{C}^{(\infty)}$ 类且

$$
\begin{aligned}
J_{\vec{f}^{-1}}(x,y) &= \begin{pmatrix} \cos\varphi & -\rho\sin\varphi \\ \sin\varphi & \rho\cos\varphi \end{pmatrix}^{-1} \\
&= \frac{1}{\rho}\begin{pmatrix} \rho\cos\varphi & \rho\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}.
\end{aligned}
$$

例 2. 已知函数 $z = z(x, y)$ 由参数方程

$$\begin{cases} x = u\cos v \\ y = u\sin v \\ z = uv \end{cases}$$

确定, 其中 $u > 0$. 试求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

解: 由于 $\frac{D(x,y)}{D(u,v)} = \begin{vmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{vmatrix} = u > 0$, 因此

存在反函数, 可将 u, v 看成是 x, y 的函数

并且

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{pmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{pmatrix}^{-1} = \frac{1}{u}\begin{pmatrix} u\cos v & u\sin v \\ -\sin v & \cos v \end{pmatrix}.$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} \cos v & \sin v \\ -\frac{1}{u}\sin v & \frac{1}{u}\cos v \end{pmatrix}.$$

于是

$$\begin{aligned}
\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x} \\
&= v\cos v + u \cdot \big(- \frac{1}{u}\sin v\big) \\
&= v\cos v - \sin v, \\
\frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y} \\
&= v\sin v + u \cdot \big(\frac{1}{u}\cos v\big) \\
&= v\sin v + \cos v.
\end{aligned}$$

例 3. 设隐函数 $u=u(x,y)$ 由方程组

$$\begin{cases} u=f(x,y,z,t) \\ g(y,z,t)=0 \\ h(z,t)=0 \end{cases}$$

确定，求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$.

解: 由题设可知，利用方程组 $\begin{cases} g(y,z,t)=0 \\ h(z,t)=0 \end{cases}$

可将 z,t 确定为 y 的函数，由此可得

$$\frac{\partial u}{\partial x}=\frac{\partial f}{\partial x}(x,y,z,t).$$

由隐函数定理可知

$$\begin{pmatrix} \frac{\mathrm{d}z}{\mathrm{d}y} \\ \frac{\mathrm{d}t}{\mathrm{d}y} \end{pmatrix} = -\begin{pmatrix} \frac{\partial g}{\partial z} & \frac{\partial g}{\partial t} \\ \frac{\partial h}{\partial z} & \frac{\partial h}{\partial t} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial g}{\partial y} \\ \frac{\partial h}{\partial y} \end{pmatrix}$$

$$= -\left(\left|\frac{\partial(g,h)}{\partial(z,t)}\right|\right)^{-1} \begin{pmatrix} \frac{\partial h}{\partial t} & -\frac{\partial g}{\partial t} \\ -\frac{\partial h}{\partial z} & \frac{\partial g}{\partial t} \end{pmatrix} \begin{pmatrix} \frac{\partial g}{\partial y} \\ 0 \end{pmatrix},$$

于是 $\frac{\mathrm{d}z}{\mathrm{d}y} = -\frac{\frac{\partial h}{\partial t}\cdot\frac{\partial g}{\partial y}}{\left|\frac{\partial(g,h)}{\partial(z,t)}\right|}$, $\frac{\mathrm{d}t}{\mathrm{d}y} = \frac{\frac{\partial h}{\partial z}\cdot\frac{\partial g}{\partial y}}{\left|\frac{\partial(g,h)}{\partial(z,t)}\right|}$, 进而可得

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}\cdot\frac{\mathrm{d}z}{\mathrm{d}y} + \frac{\partial f}{\partial t}\cdot\frac{\mathrm{d}t}{\mathrm{d}y} = \frac{\partial f}{\partial y} + \frac{\left|\frac{\partial(h,f)}{\partial(z,t)}\right|}{\left|\frac{\partial(g,h)}{\partial(z,t)}\right|}\cdot\frac{\partial g}{\partial y}.$$

同学们辛苦了！